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The Dawn of the "61% Era" for Iron Ore
铁矿石"61%时代"正在开启

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Abstract 摘要

Since the early 21st century, as a basic industrial raw material, the pricing mechanism of iron ore has undergone several key evolutions. From the withdrawal of the early annual long-term agreement negotiation mechanism to the spot index pricing model becoming the mainstream, each transformation has profoundly affected the profit distribution of the industrial chain. Today, the global iron ore market is ushering in a new round of adjustment to the pricing system: the long-dominant 62% iron grade benchmark is gradually transitioning to the 61% iron grade standard, marking the dawn of the "61% era" in the iron ore market.

The long-dominant 62% iron grade benchmark in the global iron ore market is gradually shifting to 61%. As mining advances in key Pilbara mines, PBF's iron grade has trended downward, leaving the existing 62% benchmark prone to deviations in reflecting mainstream spot. To adapt to such changes, Platts and SGX have rolled out adjustment plans, potentially pushing down far-month contract prices (starting Jan 2026) due to the lower underlying benchmark grade. Market participants should focus on Platts' "61/62% Fe Transitional Basis" during the transition and use this tool to manage pricing gap risks in intertemporal transactions for a smooth strategy shift.

Risks: Weak end-demand leads to declining profitability and increased maintenance of steel enterprises; policies such as crude steel production regulation and carbon emissions are negative for ore demand (downside risk); Policy-driven demand stimulus measures boost the overall demand in the ferrous metal industrial chain (upside risk).

自本世纪初以来，铁矿石作为基础工业原料，其定价机制经历了数次关键演变。从早期年度长协谈判机制的退出，到现货指数定价模式成为主流，每次转变都深刻影响了产业链的利润分配。如今，全球铁矿石市场正迎来新一轮定价体系调整：长期主导的62%铁品位基准正逐步向61%铁品位标准过渡，标志着铁矿石市场 “61%时代”正在开启。

随着皮尔巴拉地区核心矿区的开采进程深入，PBF 的铁品位已呈现出下滑的趋势，原有的62%基准在描述市场主流现货时易出现偏差。为适应这一现货变化，普氏和新交所均推出了调整方案，或导致远月合约（2026 年 1 月起）因标的物基准品位下调而出现价格回落。市场参与者需重点关注普氏在过渡期内发布的“61/62% Fe 过渡基差价差”，并利用该工具管理跨期交易中的定价断层风险，确保策略平稳过渡。

风险提示：终端需求疲弱导致钢企盈利下降、检修增加；粗钢产量调控、碳排放等政策利空矿石需求（下行风险）；政策端释放刺激需求利好，提振黑色产业链整体需求（上行风险）。

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Content 目录

Charts 图表

Quality adjustment of PBF is the core driver of benchmark restructuring, and the effectiveness of the iron ore pricing benchmark depends on its representativeness. **As the mining progress of core mining areas in the Pilbara region advances in depth, the iron grade of PBF has shown a downward trend.** This makes the original 62% benchmark prone to deviations when describing the mainstream spot goods in the market, thereby driving the revision of this index rule. At the same time, the new standard has moderately relaxed and optimized the thresholds of impurity indicators to more inclusively reflect the real mineral characteristics of current mainstream resources.

To adapt to this change in spot goods, major information agencies and exchanges have all launched clear adjustment plans:

Platts Rule Change: Starting from January 2, 2026, the iron content benchmark of IODEX will be lowered from 62% to 61%, the silicon dioxide (SiO_2) benchmark will be adjusted from 4.0% to 4.5%, and the aluminum oxide (Al_2O_3) benchmark will be adjusted from 2.25% to 2.5%. This adjustment aims to reduce the error generated when converting the current mainstream 61% grade ore to the 62% benchmark, and improve the guiding significance of the index for the spot market.

SGX Contract Adjustment: It has adopted an automatic anchoring mechanism of "new-old segmentation". Contracts expiring before January 2026 will continue to anchor the 62% index; while contracts expiring in and after January 2026 will automatically anchor the new 61% IODEX.

This adjustment will directly lead to a price decline of far-month contracts (starting from January 2026) due to the downgrade of the underlying benchmark grade. This is not caused by the deterioration of supply and demand, but a price change brought by the alteration of the underlying asset. Market participants need to focus on the **"61/62% Fe Transitional Basis Spread"** released by Platts during the transition period, and use this tool to manage the pricing gap risk in intertemporal transactions, so as to ensure the smooth transition of trading strategies during the migration to updated specifications.

1. The "61% Era" of Iron Ore is Dawning

Since the early 21st century, as a basic industrial raw material, the pricing mechanism of iron ore has undergone several key evolutions. From the withdrawal of the early annual long-term contract negotiation mechanism to the mainstream spot index pricing model, each transformation has profoundly affected the profit distribution across the industrial chain. Today, the global iron ore market is ushering in a new round of pricing system adjustment: the long-dominant 62% Fe benchmark is shifting to a 61% Fe index.

This change is not merely a simple downgrade of iron grade, but a result of the shifting trend of resource endowments. As an important pricing basis for the market, S&P Global Commodity Insights (hereinafter referred to as "Platts") has announced that its core iron ore price index IODEX will officially adopt 61% iron content as the benchmark starting from January 2, 2026. In the same period, the Singapore Exchange (SGX) has also made corresponding adjustments to the specifications of its iron ore derivatives contracts. These measures mark that the 62% benchmark system that has prevailed in the iron ore market for many years is about to change, and the "61% Era" is dawning.

2. Mainstream Mines' Grade Reduction Drives Index Benchmark Adjustment

The effectiveness of the iron ore pricing benchmark is founded on its representativeness. For a long time, the reason why the 62% Fe index has become the benchmark is that it accurately corresponds to the standardized product with the best liquidity and largest trading volume in the market—the Australian medium-grade fine ore represented by Rio Tinto's Pilbara Blend Fines (PBF). However, with the increase of mining years, the iron grade of core mining areas in the Pilbara region has shown an irreversible downward trend. This is not only the case for Australian mines, but Vale in Brazil has also experienced a similar grade decline. This phenomenon reflects that mainstream global mines are generally facing the problem of grade reduction, and has driven corresponding adjustments to relevant representative index benchmarks.

2.1 Grade Reduction of Rio Tinto's Flagship Product

As a benchmark product in the global iron ore market, PBF has long provided a key reference for market prices due to its stable quality and supply scale. However, production data since last year indicates that Rio Tinto has been unable to maintain the iron content of PBF at the 62% level consistently. According to analysis of market tracking data and Rio Tinto's official reports, the actual iron content of PBF has systematically transitioned to the range of 61% and even lower.

2024 Structural Dilemma: PBF Output Decline and SP10's Substitution.

Data in 2024 demonstrated the restraining effect of "grade maintenance" on output. Due to the depletion of core mining areas such as Yandicoogina and Paraburdoo, it has become extremely difficult to maintain high grades. Throughout 2024, Rio Tinto's PBF shipment volume was only 97.8 million tons, a significant year-on-year decrease of 7%. In some periods (such as the first half of the year), the sales decline of PBF even reached 15%. To maintain the stability of the overall shipment target, Rio Tinto had to convert low-grade ores excluded from the PBF series into SP10 (SP10 Fines/Lump) for sale. Data shows that in 2024, SP10 lump ore shipments surged by 86%, and SP10 fine ore shipments surged by 17%, totaling 63.8 million tons. Although this total volume maintenance strategy preserved its market share, it also reduced the company's average selling price and led to a decline in its EBITDA to a certain extent.

Official Confirmation of Grade Reduction: In 2025, Rio Tinto confirmed to the market that the iron grade index of its PBF product has been adjusted to approximately 60.8%, while its lump ore product (Pilbara Blend Lump) has been lowered to 61.6%.

2.2 Sharp Contraction of Vale's High-Grade IOCJ Output and Overall Quality Decline

The Carajás mining area of Vale located in Pará State, northern Brazil, has long been regarded as the benchmark for global iron ore quality. Its flagship product IOCJ (Iron Ore Carajás), with an iron content of approximately 65% and extremely low silicon-aluminum impurities, enjoys a significant market premium. However, production data from 2024 to the present shows that this "myth" is facing severe realistic challenges.

In its Q3 2025 production report, Vale disclosed significant anomalies in IOCJ sales data. The sales volume of IOCJ in this quarter was only 5.67 million tons, a sharp year-on-year decrease of 51.6%. In the first three quarters of 2025, the sales volume of IOCJ was 16.67 million tons, a year-on-year decrease of 51.4%. This decline is extremely rare in Vale's recent history, reflecting the tightening of high-grade ore supply.

The core reason for the limited IOCJ output lies in the delayed permit approval for the Serra Norte mining area in the northern system. Serra Norte is the main mining section for producing IOCJ. As existing pits gradually mature, the grade fluctuation of raw ore has intensified. To maintain the stringent 65% Fe standard of IOCJ, Vale needs to continuously develop new high-grade ore bodies for ore blending. However, the lag in environmental permit approval has restricted the stripping and mining of new pits. Faced with the inevitable decline in IOCJ output, Vale has adopted an extremely flexible product portfolio optimization strategy and launched a new product—Mid-Grade Carajás (~63% Fe).

Although Mid-Grade Carajás has compensated for the reduction of IOCJ to a certain extent, the overall iron content has declined. Vale's average iron grade dropped from 62.4% in Q4 2024 to 61.7%, while the silicon content rose from 5.8% to 7.0%, showing an overall downward trend in quality.

2.3 FMG and Other Australian Mines Also Show Grade Reduction Trends

FMG plans to phase out West Pilbara Fines (WPF) gradually and introduce new low-grade products to fill the gap. In its Q1 FY26 production report (September 2025 quarter), FMG officially announced a major adjustment to its product portfolio: it plans to phase out its flagship medium-grade product—WPF (60% Fe)—gradually in FY27. To fill the output gap left by WPF's withdrawal, FMG announced that it will introduce a new 55% Fe product (expected to account for 5-6% of total hematite shipments). At the same time, the company has pinned all its hopes for high-grade ore on the Iron Bridge project, but the Iron Bridge project is in the capacity ramp-up phase, with limited short-term output release.

Data from mid-sized miners such as MinRes (Mineral Resources) is more intuitive. The average grade of products from its main Pilbara Hub mining area has gradually declined from 58.3% (Q2 FY24) to 56.9% (Q1 FY26).

2.4 Structural Rise in Silicon and Aluminum Impurity Content

In addition to the decline in iron content, the rise in impurity content is a more intractable challenge. As the mining depth of ore bodies in the Pilbara region increases, the content of mixed surrounding rock and gangue components has increased, directly leading to a rise in the content of silicon dioxide (SiO_2) and aluminum oxide (Al_2O_3).

Silica (SiO_2): As the main component of acidic gangue, the increase in silicon content directly increases the slag volume during ironmaking. Platts' original 4.0% silicon benchmark can no longer cover the actual level of current mainstream Australian fine ore. The silicon content of the new PBF will rise from around 3.6% in the past to 4.3%.

Aluminum Oxide (Al_2O_3): Aluminum oxide has a significant negative impact on the fluidity and desulfurization capacity of blast furnace slag, and is the most sensitive harmful impurity for steel mills. The aluminum content of the new PBF will rise from around 2.3% in the past to 2.5%.

3. Index Adjustments by Platts and SGX

This industry-wide quality decline has led to an awkward market reality: the benchmark index describes an "idealized" product, while actual transactions are based on "discounted" products. When most mainstream spot goods in the market need to undergo substantial standardization calculations to be converted to the 62% benchmark, the significance of the index as the pricing benchmark is greatly weakened. This requires the index to adjust with the times to meet current needs.

3.1 Platts IODEX Index Change

As the authoritative institution for global iron ore pricing, the methodological adjustment of Platts' IODEX index is the core of this round of index adjustments. According to the announcement it released, this adjustment not only involves numerical changes but also includes a set of complex transition mechanisms.

(1) Specific Indicator Changes to Platts IODEX

Starting from January 2, 2026, Platts IODEX and a series of its derived price assessments will implement new quality standards. This change targets not only iron content but also systematically relaxes impurity thresholds.

Table 1: Platts IODEX Indicator Change Comparison

Key Indicators	Old Standard (Before 2026)	New Standard (Effective from January 2, 2026)	Change
Iron Content (Fe)	**62.00%**	**61.00%**	**-1.00%**
Silicon Dioxide (SiO_2)	4.00%	4.50%	**+0.50%**
Aluminum Oxide (Al_2O_3)	2.25%	2.50%	**+0.25%**
Phosphorus (P)	0.09%	0.10%	**+0.01%**
Sulfur (S)	0.02%	0.02%	0.00%
Moisture (H_2O)	8.00%	8.00%	0.00%

Source: S&P Global Energy, CITIC Futures Research Institute

Iron content (Fe) has been significantly lowered from 62% to 61%. This acknowledges the reality of declining grades in mainstream medium-grade iron ore globally, reduces the error when converting 61% ore to the 62% benchmark, and enhances the index's representativeness.

The benchmark for silicon dioxide (SiO_2) has been significantly relaxed. Platts' original 4.0% silicon benchmark can no longer cover the actual level of current mainstream Australian fine ore. The silicon content of the new PBF will rise from around 3.6% in the past to 4.3%.

The benchmark for aluminum oxide (Al_2O_3) has been slightly increased. Aluminum oxide is the most difficult impurity to handle in blast furnace operations. Raising the benchmark means that buyers must either accept a higher aluminum load or pay an additional premium to purchase low-aluminum ore.

The benchmark for phosphorus (P) has been minorly adjusted. As the proportion of mixed-in high-phosphorus ore increases, relaxing the phosphorus content helps improve the index's inclusivity.

The benchmark for sulfur (S) remains unchanged. Environmental regulations on sulfur emissions are still strict, so no adjustment has been made.

The benchmark for moisture (H_2O) remains unchanged. The benchmark used for converting between wet tons and dry tons stays stable.

(2) Transition Period Arrangements

To address the pricing gap issue in intertemporal transactions from late 2025 to early 2026, Platts has designed a sophisticated transition mechanism. Since the index reflects spot prices, and spot transactions often feature forward shipment dates within 14 to 56 days, by the end of 2025, the market will trade both near-term cargoes meeting the 62% standard and forward (2026 CIF) cargoes meeting the 61% standard simultaneously.

From July 1 to December 31, Platts will conduct an independent assessment of the IODEX spread and provide daily reference prices consistent with the updated IODEX quality specifications during the transition period. This assessment is based on transactions, bids, offers, and other market information reported to Platts, reflecting the spread between the current 62% Fe specification and the updated 61% Fe specification. Under the independent assessment mechanism for the basis spread, Platts may standardize market information to account for value differences caused by physical and metallurgical properties, quality stability, and other factors.

From January 2, 2026, to December 31, 2027, Platts will begin daily publication of the 61/62% Fe Transitional Basis Spread (daily ticker: FIOTB00; monthly average ticker: FIOTB03) to support the market's smooth migration to the updated IODEX quality specifications. This indicator will be released as a positive value, and its calculation is based on the IODEX price, and Platts' daily assessed prices for silicon, aluminum, and phosphorus, combined with adjustments for iron grade. The formula is as follows: 61/62% Fe Transitional Basis Spread = Updated IODEX basis 61% Fe/61*62 + Total value-in-use difference for impurity contents between 61% Fe- and 62% Fe-basis specifications – Updated IODEX basis 61% Fe

Calculations for the total value-in-use difference for impurity contents between 61% Fe- and 62% Fe-basis specifications will be as follows:

Table 2: Calculations for the Total value-in-use Difference for Impurity Contents

Key Indicators	New Standard	Old Standard	Differential Assessment	Formula
Silicon Dioxide (SiO_2)	4.50%	4.00%	Iron Ore Silica Differential per 1% with 3-4.5% (IOALF00)	IOALF00 * (4.5-4)
Aluminum Oxide (Al_2O_3)	2.50%	2.25%	Iron Ore Alumina Differential per 1% with 1-2.5% (IOADF10)	IOADF10 * (2.5-2.25)
Phosphorus (P)	0.10%	0.09%	Iron Ore Phosphorus Differential per 0.01% with 0.09-0.10% (IOPPR00)	IOPPR00 * (0.1-0.09)/0.01

Source: S&P Global Energy, CITIC Futures Research Institute

3.2 Adjustments to Derivatives Contracts of the Singapore Exchange (SGX)

Changes in the global iron ore spot market have also directly driven adjustments to the underlying assets of the Singapore Exchange. To adapt to the new dynamics of the spot market, the exchange has optimized the relevant derivatives contracts to enhance the market adaptability and liquidity of the contracts, and better meet the trading and risk management needs of different types of market participants.

SGX has adopted a more convenient method to handle the benchmark change, namely: contracts expiring before January 2026 will continue to anchor the TSI (62%) index, and their cash settlement price will use the monthly arithmetic average of the 62% index. Contracts expiring in and after January 2026 will automatically anchor the new 61% Fe IODEX, and their cash settlement price will use the monthly arithmetic average of the 61% index.

Since the value of 61% grade ore is lower than that of 62% grade ore, the price of the January 2026 contract will be significantly lower than that of the December 2025 contract. On SGX's forward curve, there will be a substantial price movement between the December 2025 contract and the January 2026 contract. However, this is not caused by supply and demand factors, but by the change of the underlying index.

Minor Adjustments to Minimum Price Fluctuation and Multiplier: To maintain the consistency between the contract value and the underlying asset, SGX has made minor adjustments to the multiplier logic of some contracts (such as lump ore premium futures): 1 contract = 100 dry metric tons × iron grade (62% for contracts expiring before January 2026, 61% for contracts expiring in and after January 2026). This has led to a change in the minimum price movement value per contract. With a minimum price tick size per dry metric ton of $0.0001, the value of the minimum price tick for contracts expiring before January 2026 is $0.62, and that for contracts expiring in and after January 2026 is $0.61).

This adjustment will directly lead to a price decline of far-month contracts (starting from January 2026) due to the downward adjustment of the benchmark grade of the underlying asset. This is not caused by the deterioration of supply and demand, but a price change brought about by the change of the underlying asset. Market participants need to focus on the "61/62% Fe Transition Basis Spread" released by Platts during the transition period, and use this tool to manage the pricing gap risk in intertemporal transactions, so as to ensure the smooth transition of strategies during the index switch.

Appendix: Chinese Version

PBF 品质调整是基准重构的核心驱动，铁矿石定价基准的有效性依赖于其代表性。**随着皮尔巴拉地区核心矿区的开采进程深入，PBF 的铁品位已呈现出下滑的趋势。**这使得原有的 62%基准在描述市场主流现货时易出现偏差，进而推动了此次指数规则的修订。同时，新标准对杂质指标的阈值进行了适度放宽与优化，以更包容地反映当前主流资源的真实矿物特征。

为适应这一现货变化，主要资讯机构与交易所均推出了明确的调整方案：

普氏（Platts）规则变更：自 2026 年 1 月 2 日起，IODEX 基准铁含量由 62%下调至 61%，二氧化硅（SiO_2）基准由 4.0%调整至 4.5%，氧化铝（Al_2O_3）由 2.25%调整至 2.5%。这一调整旨在减少将当前主流 61%品位矿石折算为 62%基准时产生的误差，提升指数对现货市场的指导意义。

新交所（SGX）合约调整：采取了“新老划断”的自动锚定机制。2026 年 1 月之前到期的合约继续锚定 62%指数；而 2026 年 1 月及之后到期的合约将自动锚定新的 61% IODEX。

此次调整将直接导致远月合约（2026 年 1 月起）因标的物基准品位下调而出现价格回落，这并非供需恶化所致，而是标的变更带来的价格变化。市场参与者需重点关注普氏在过渡期内发布的“61/62% Fe 过渡基差价差”，并利用该工具管理跨期交易中的定价断层风险，确保策略在指数切换中的平稳过渡。

1. 铁矿石“61%时代”正在开启

自本世纪初以来，铁矿石作为基础工业原料，其定价机制经历了数次关键演变。从早期年度长协谈判机制的退出，到现货指数定价模式成为主流，每次转变都深刻影响了产业链的利润分配。如今，全球铁矿石市场正迎来新一轮定价体系调整：长期主导的 62%铁品位基准正逐步向 61%铁品位标准过渡。

这一变化不仅是品位的简单下调，更是资源禀赋趋势变迁的结果。作为市场重要定价依据，标普全球大宗商品（S&P Global Commodity Insights，下称“普氏”）已宣布，其核心铁矿石价格指数 IODEX 将自 2026 年 1 月 2 日起正式采用 61%铁含量作为基准。同期，新加坡交易所（SGX）也对旗下铁矿石衍生品合约规格进行了相应调整。这些举措标志着铁矿石市场持续多年的 62%基准体系即将改变，“61%时代”正在开启。

2. 主流矿山降品驱动指数基准调整

铁矿石定价基准的有效性建立在其代表性之上。长期以来，62%Fe 指数之所以成为基准，是因为其精准对应了市场上流动性最好、成交量最大的标准化产品——以力拓的皮尔巴拉混合粉（Pilbara Blend Fines, PBF）为代表的澳洲中品粉矿。然而，随着开采年限的增加，皮尔巴拉地区核心矿区的铁品位正呈现不可逆的下降趋势。不仅澳大利亚矿山如此，巴西淡水河谷也出现了类似的品位下滑情况。这一现象反映出全球主流矿山普遍面临品位下降的问题，并推动相关代表性指数基准作出相应调整。

2.1 力拓旗舰产品 PBF 铁品位下调

作为全球铁矿石市场的基准产品，PBF 以其品质稳定与供应规模，长期为市场价格提供关键参照。然而，自去年以来的生产数据表明，力拓已难以持续将 PBF 的铁含量维持在 62% 的水平。根据市场追踪数据及力拓官方报告分析，PBF 的实际铁含量已系统性地向 61%乃至更低的区间过渡。

2024 年的结构性困境：PBF 产量下降与 SP10 的补位。2024 年的数据展示了“保品位”对产量的抑制作用。由于 Yandicoogina 和 Paraburdoo 等核心矿区的枯竭，维持高品位变得极度困难。2024 年全年，力拓 PBF 的发货量仅为 9780 万吨，同比大幅下降 7%。在部分时段（如上半年），PBF 的销售降幅甚至达到了 15%。为了维持整体发货目标的稳定，力拓只能将被剔除出 PBF 序列的低品位矿石转化为 SP10（SP10 Fines/Lump）进行销售。数据显示，2024 年 SP10 块发运激增 86%、SP10 粉发运量激增 17%，合计达到 6380 万吨。这种维持总量的策略虽然保住了市场份额，但也降低了公司平均销售价格，一定程度上导致其 EBITDA 下降。

品位下调的官方确认：力拓已在 2025 年向市场确认，其 PBF 产品的铁品位指标已调整为约 60.8%，而其块矿产品（Pilbara Blend Lump）下调至 61.6%。

2.2 Vale 高品 IOCJ 产量大幅收缩，整体品质下行

淡水河谷位于巴西北部帕拉州的卡拉加斯矿区长期以来被视为全球铁矿石质量的标杆。其主力产品 IOCJ（Iron Ore Caraj á s）以约 65%的铁含量和极低的硅铝杂质，享有显著的市场溢价。然而，2024 年至今的生产数据显示，这一“神话”正在遭遇现实的严峻挑战。

在 2025 年第三季度的生产报告中，淡水河谷披露了 IOCJ 销量数据的显著异动。该季度 IOCJ 的销售量仅为 567 万吨，与 2024 年同期相比大幅下滑了 51.6%。2025 年前三个季度 IOCJ 销售量 1667 万吨，同比下滑 51.4%。这一降幅在淡水河谷的近期历史中极为罕见，反映了高品位矿石供应的紧缩。

造成 IOCJ 产量受限的核心原因在于北部系统 Serra Norte 矿区的许可审批延迟。Serra Norte 是生产 IOCJ 的主要矿段，随着现有矿坑的逐渐成熟，原矿的品位波动加剧。为了维持 IOCJ 65%Fe 的严苛标准，淡水河谷需要不断开发新的高品位矿体以进行配矿。然而，环境许可的审批滞后限制了新

矿坑的剥离与开采。面对 IOCJ 产量的不可避免的下滑，淡水河谷采取了极其灵活的产品组合优化策略，推出了新产品——Mid-Grade Carajás(~63% Fe)。

虽然 Mid-Grade Carajás 在一定程度上弥补了 IOCJ 的减量，但整体铁元素含量有所下滑。淡水河谷平均铁品位由 2024 年四季度的 62.4%降至 61.7%，硅含量则由 5.8%上升至 7.0%，整体呈现出品质下行的趋势。

2.3 FMG 和澳洲其他矿山同样有降品趋势

FMG 计划逐步淘汰 West Pilbara Fines（WPF），引入新的低品位产品填补缺口。在 2025 年 9 月季度生产报告(Q1 FY26)中，FMG 正式宣布了对其产品组合的重大调整：计划在 2027 财年(FY27)逐步淘汰其旗舰中品位产品——WPF(60% Fe)。为了填补 WPF 退出后的产量缺口，FMG 宣布将引入一种新的 55% Fe 产品（预计将占赤铁矿总发货量的 5-6%）。与此同时，公司将全部的高品位希望寄托于 Iron Bridge 项目，但 Iron Bridge 项目在产能爬坡中，短期释放量有限。

MinRes（Mineral Resources）等中型矿商的数据更为直观，其主要矿区 Pilbara Hub 产品的平均品位从 58.3%(Q2 FY24) 逐步下滑至 56.9%(Q1FY26)。

2.4 硅、铝杂质含量的结构性上升

除铁元素含量下降外，杂质含量的上升是更为棘手的挑战。皮尔巴拉地区的矿体随着开采深度的增加，混入的围岩和脉石成分增多，直接导致二氧化硅（SiO_2）和氧化铝（Al_2O_3）含量上升。

二氧化硅（Silica）：作为酸性脉石的主要成分，硅含量的上升直接增加了炼铁过程中的渣量。普氏原有的 4.0%硅基准已无法覆盖当前主流澳洲粉矿的实际水平。新 PBF 的硅含量将从过去的 3.6%左右攀升至 4.3%。

氧化铝（Al_2O_3）：氧化铝对高炉渣的流动性和脱硫能力有显著负面影响，是钢厂最为敏感的有害杂质。新 PBF 的铝含量将从过去的 2.3%左右攀升至 2.5%。

3. 普氏与新交所指数调整

这种全行业的品质下降导致了一个尴尬的市场现实：基准指数描述的是一个“理想化”的产品，而现实交易的却是“打折”的产品。当市场上多数的主流现货都需要经过大幅度的标准化计算才能折算到 62%基准时，指数的定价基准意义就被大幅弱化。这就需要指数与时俱进进行调整，以满足当下的需求。

3.1 普氏（Platts）IODEX 指数变更

普氏作为全球铁矿石定价的权威机构，其 IODEX 指数的方法论调整是此次指数调整的核心。根据其发布的公告，此次调整不仅涉及数值的变更，更包含了一套复杂的过渡机制。

（1）普氏 IODEX 具体指标变更

自 2026 年 1 月 2 日起，普氏 IODEX 及其衍生的一系列价格评估将执行新的质量标准。这一变更不仅针对铁含量，还系统性地放宽了杂质阈值

图表1：普氏 IODEX 指标变更对照表

关键指标	旧标准（2026 年之前）	新标准（2026 年 1 月 2 日生效）	变化
铁含量(Fe)	62.00%	61.00%	-1.00%
二氧化硅(SiO_2)	4.00%	4.50%	+0.50%
氧化铝(Al_2O_3)	2.25%	2.50%	+0.25%
磷(P)	0.09%	0.10%	+0.01%
硫(S)	0.02%	0.02%	0.00%
水(H_2O)	8.00%	8.00%	0.00%

资料来源：S&P Global Energy 中信期货研究所

铁含量（Fe）由 62%显著下移至 61%。承认了全球主流中品矿品位下降的现实，减少了将 61%矿石折算为 62%时的误差，提升指数代表性。

二氧化硅（SiO_2）大幅放宽。普氏原有的 4.0%硅基准已无法覆盖当前主流澳洲粉矿的实际水平。新 PBF 的硅含量将从过去的 3.6%左右攀升至 4.3%。

氧化铝（Al_2O_3）小幅上调。氧化铝是高炉操作中最难处理的杂质。提高基准意味着买方必须接受更高的铝负荷，或支付额外溢价购买低铝矿。

磷（P）微调。随着高磷矿混入比例增加，放宽磷含量有助于提高指数的包容性。

硫（S）维持不变。环保法规对硫排放的限制依然严格，因此未做调整。

水(H_2O)维持不变。用于湿吨与干吨换算的基准保持稳定。

（2）过渡期安排

为了解决 2025 年底至 2026 年初跨期交易的定价断层问题，普氏设计了一个精密的过渡机制。由于指数反映的是现货价格，而现货交易往往包含 14-56 天后的远期船期，因此在 2025 年末，市场将同时交易 62% 标准的近期货物和 61% 标准的远期（2026 年到岸）货物。

在 7 月 1 日至 12 月 31 日期间普氏将对 IODEX 价差进行独立评估，并在过渡期内提供与更新后 IODEX 质量规范一致的每日参考价格。该评估基于向普氏报告的成交、买盘、卖盘及其他市场信息，反映当前 62% Fe 规格与更新后 61% Fe 规格之间的价差。在基差价差采用独立评估的机制下，普氏可能对市场信息进行标准化处理，以考虑因物理与冶金性质、质量稳定性及其他因素带来的价值差异。

自 2026 年 1 月 2 日起至 2027 年 12 月 31 日，普氏将开始每日发布 61/62% Fe 过渡基差价差（每日代码 FIOTB00；月均代码 FIOTB03），以支持市场向更新后的 IODEX 质量规范平稳过渡。该指标将以正值形式发布，其计算基于 IODEX 价格，以及普氏每日针对硅、铝、磷的评估价，并结合铁品位调整，计算公式如下：

61/62% Fe 过渡基差价差 =((更新后 IODEX 基准 61%Fe）/ 61 × 62)+(61%Fe 与 62%Fe 基准规格之间杂质含量的价差总和）－（更新后 IODEX 基准 61% Fe）

其中，61% Fe 与 62% Fe 基准规格之间杂质含量的价差的计算方式如下：

图表2：杂质价差评估

关键指标	新标准	旧标准	杂质指标评估价	计算公式
二氧化硅(SiO_2)	4.50%	4.00%	每 1%硅含量差价，适用于 3－4.5%区间（IOALF00）	IOALF00 * (4.5-4)
氧化铝(Al_2O_3)	2.50%	2.25%	每 1%铝含量差价，适用于 1-2.5% (IOADF10)	IOADF10 * (2.5-2.25)
磷(P)	0.10%	0.09%	每 0.01%磷含量差价，适用于 0.09-0.10% (IOPPR00)	IOPPR00 * (0.1-0.09)/0.01

资料来源：S&P Global Energy 中信期货研究所

3.2 新加坡交易所（SGX）衍生品合约的调整

全球铁矿石现货市场的变化，也直接推动了新加坡交易所标的的调整。为适应现货市场的新动态，交易所对相关衍生品合约进行了优化，以提升合约的市场适应性和流动性，更好地满足不同类型市场参与者的交易与风险管理需求。

SGX 采取了一种更为便捷的方式来处理基准变更，即：2026 年 1 月之前到期的合约继续锚定 TSI (62%)指数，其现金结算价采用 62% 指数的月度算术平均值。2026 年 1 月及之后到期的合约自动锚定新的 61% Fe IODEX，其现金结算价采用 61% 指数的月度算术平均值。

由于 61%矿石的价值低于 62%矿石，2026 年 1 月合约的价格将显著低于 2025 年 12 月合约。在 SGX 的远期曲线上，2025 年 12 月合约和 2026 年 1 月合约之间会有大幅的价格变动。但这并非供需因素所致，而是标的指数变更引起的。

最小价格波幅与乘数的微调：为了保持合约价值与标的物的一致性，SGX 对部分合约（如块矿溢价期货）的乘数逻辑进行了微调：1 手合约 =100 千吨 × 铁品位(2026 年 1 月前到期合约为 62%，2026 年 1 月及以后到期合约为 61%），由此导致的最小价格波幅也发生了变化：每千吨单位 0.0001 美元（2026 年 1 月前到期合约最小变动价位价值=0.62 美元；2026 年 1 月及以后到期合约最小变动价位价值=0.61 美元）。

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